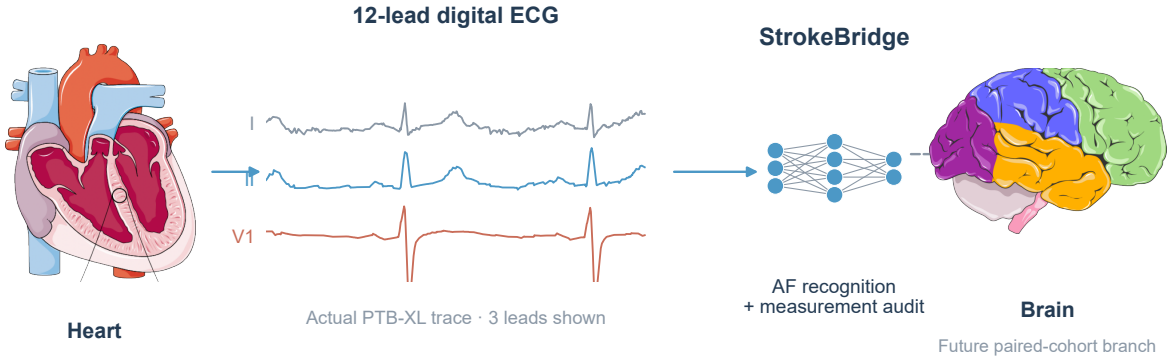
a

From cardiac electrical signals to brain–heart research



b

Patient-isolated development

	ECGs
Train — 12,743 patients · 1047 AF ECGs	14,709
Select — 1,825 patients · 151 AF ECGs	2,101
Calibrate — 1,870 patients · 149 AF ECGs	2,097
Evaluate — 1,822 patients · 150 AF ECGs	2,095

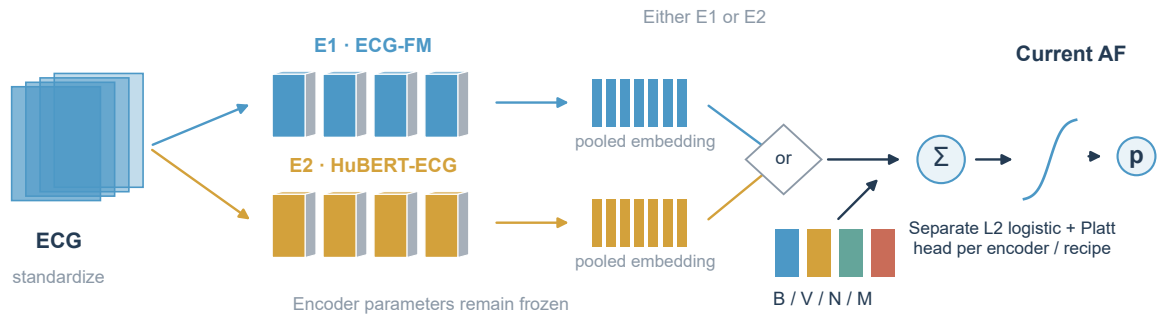
c

Explicitly partitioned information groups

- **B14** — Clinical / RR / quality
- **V31** — Candidate visibility
- **N6** — Counts / matching
- **M25** — Candidate morphology

d

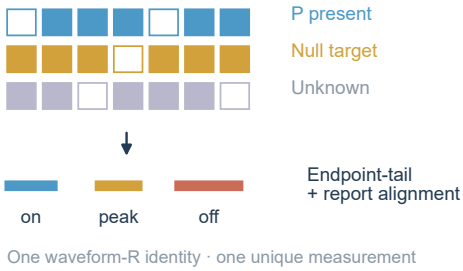
Two frozen encoders, one information-separated prediction protocol



e

Patient-complete atrial supervision

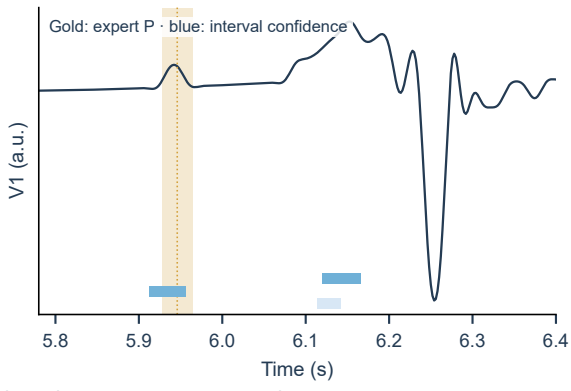
Lead II / V1 × all labelled heartbeats



One waveform-R identity · one unique measurement

f

ECG-aligned candidate confidence



Current evidence: AF identification and method auditing. Future AF / stroke risk requires a separate outcome cohort.